rectangle. You can use the function *pspolygon* to draw polygons with any number of sides. The code fragment:

```
[linecolor=red,linewidth=8pt]
```

…sets the colour of the line to red and sets the line width to eight points. As a result, the red rectangle will be drawn with thick lines, whereas the blue rectangle will be drawn with thin lines.

Now that we are familiar with the code, it is time to execute and view the output document. As mentioned earlier, we will discuss two methods to obtain output from a LaTeX file containing PSTricks code. The first method uses a LaTeX editor called Texmaker — or any other LaTeX editor for that matter. Open the LaTeX file *pstricks1.tex* in Texmaker and from the menu, choose *Tools>LaTeX* to execute *pstricks1.tex* to obtain the output file *pstricks1.dvi*. Even if you are using another LaTeX editor, you will still be able to use the tool LaTeX to process the LaTeX file — the reason is simple: any LaTeX editor will at least offer LaTeX as a tool if nothing more.

We have already discussed files with extension *dvi* (device independent file format) in the previous article on PGF/TikZ. We are all in need of PDF outputs but converting a DVI file containing PostScript directly to PDF will lead to errors. So, the trick is to convert the DVI file first to a PostScript file with extension *ps* — for which we use a tool called dvips — and then convert this PostScript file to a PDF file — for which we use a tool called ps2pdf. All you need is Texmaker.

To convert the DVI output file to PS format, choose and execute *Tools>Dvi->PS*. Now, to convert this output file in PS format to PDF format, choose and execute *Tools>PS->PDF*. For the remaining part of this article, we will only produce DVI outputs but, if necessary, the above two steps can be repeated to obtain an output file in PDF format. After the execution of these three steps, if you check the directory containing the LaTeX script *pstricks1.tex*, you will find three different output files, all of which contain the image shown in Figure 1. The output files are *pstricks1.dvi*, *pstricks1.ps* and *pstricks1.pdf*. But make sure that you have a document viewer capable of displaying all these formats and for this purpose, I recommend Okular.

The other method to obtain the output file is to use the commands *latex*, *dvips* and *ps2pdf* directly on the terminal. Open a terminal in the directory containing the file *pstricks1.tex* and execute the command '*latex pstricks2.tex*' to obtain the output file *pstricks1.dvi*. Execute the command '*dvips pstricks2.dvi*' to convert the file *pstricks1.dvi* to the output file *pstricks1.ps* in PostScript format. Execute the command '*ps2pdf pstricks2.ps*' to convert the file *pstricks1.ps* to the output file *pstricks1.pdf* in PDF format.

Now let us understand the function *pspolygon* better by

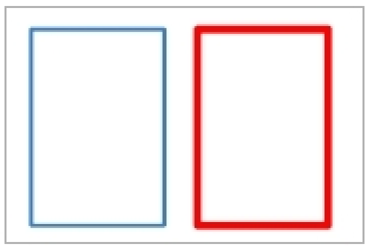

Figure 1: Two rectangles drawn with PSTricks

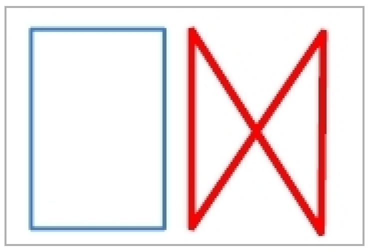

Figure 2: Output of *pstricks2.tex*

making some slight modifications in the code of *pstricks1.tex*. What will be the output if the line of code:

```
\pspolygon[linecolor=red, linewidth=8pt](6,1)
(10,1)(10,7)(6,7)
```

in *pstricks1.tex* is replaced with the line of code:

```
\pspolygon[linecolor=red,linewidth=8pt](6,1)
(10,7)(10,1)(6,7)
```

On execution with Texmaker or in the terminal using commands, this modified file called *pstricks2.tex* will produce the output shown in Figure 2. But why? Well, PSTricks has just followed your instructions — that's all. A line is drawn from point (6,1) to point (10,7) – the forward slanting line going upward; followed by a line from point (10,7) to point (10,1) — a vertical line going downward; followed by a line from point (10,1) to point (6,7) — the backward slanting line going upward. And finally, a line from point (6,7) to point (6,1) — a vertical line going downward. So, the script *pstricks2.tex* results in the image shown in Figure 2.

Drawing circles with PSTricks

Let us try to draw a circle using PSTricks. Consider the LaTeX script *pstricks3.tex* shown below which draws a circle and a square.

```
\documentclass{article}
\usepackage{pstricks}
\begin{document}
    \begin{pspicture}(15,15)
      \pspolygon[linecolor=red,fillstyle=solid,fillcolor=yell
ow](1,1)(3,1)(3,3)(1,3)
      \pscircle[linecolor=blue,fillstyle=solid,fillcolor=gre
en](2,2){1}
    \end{pspicture}
\end{document}
```

Now let us go through the code in detail. In this script, all the lines of code are familiar to us except for two lines which have been newly introduced. The line of code to draw the square is:

```
\pspolygon[linecolor=red,fillstyle=solid,fillcolor=yellow](1,1)
(3,1)(3,3)(1,3)
```

This line draws a square that has sides of two units. The code fragment:

```
[linecolor=red,fillstyle=solid,fillcolor=yellow]
```